

Systems of Equations — Number of Solutions

Full Solutions & Explanations

Difficulty Mix: Easy (Q1–4) | Medium (Q5–9) | Hard (Q10–12) · Total: 12 Questions

Q1. Correct Answer: A [EASY]

STEP-BY-STEP SOLUTION:

Both lines have slope 3 but different y-intercepts ($5 \neq -2$).

Parallel lines with different y-intercepts never intersect.

Number of solutions: 0. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) ✓ CORRECT — CORRECT. Same slope (3), different y-intercepts → parallel → no solution.

B) One solution requires different slopes.

C) Two lines can intersect at most once.

D) Requires both slope AND y-intercept to match.

■ **TRAP ALERT:** Same slope = parallel (unless also same intercept). Here $5 \neq -2$, so the lines are distinct and parallel.

■ **KEY INSIGHT:** For $y=mx+b$ vs $y=mx+b$: $b \neq b \rightarrow 0$ solutions. $b = b \rightarrow \infty$ solutions. Different $m \rightarrow 1$ solution.

■ **FAST METHOD:** Same slope (3), different intercepts (5 vs -2) → parallel → 0 solutions.

Q2. Correct Answer: C [EASY]

STEP-BY-STEP SOLUTION:

Infinitely many solutions means the equations represent the SAME line.

Check C: $y=2x+1$ and $4y=8x+4$.

Divide $4y=8x+4$ by 4: $y=2x+1$. ✓ Identical to the first equation.

Answer: C. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Same slope, different intercepts → parallel → 0 solutions.

B) Different slopes → intersect at exactly 1 point.

C) ✓ CORRECT — CORRECT. $4y=8x+4$ simplifies to $y=2x+1$ — same line → ∞ solutions.

D) Different slopes → 1 solution.

■ **TRAP ALERT:** Always simplify before comparing. $4y=8x+4$ must be divided by 4 to reveal it's the same line.

■ **KEY INSIGHT:** ∞ solutions $\leftrightarrow$ one equation is a scalar multiple of the other (same line).

■ **FAST METHOD:** Divide equation 2 by 4 $\rightarrow y=2x+1$. Identical to equation 1 $\rightarrow \infty$ solutions.

Q3. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

Line 1: $y = 2x + 1$ (slope = 2).

Line 2: $y = -x + 7$ (slope = -1).

Since the slopes are different ($2 \neq -1$), the lines are not parallel.

Two non-parallel lines intersect at exactly 1 point $\rightarrow$ 1 solution. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Zero solutions requires parallel lines (same slope). Here slopes differ.

B) ✓ CORRECT — CORRECT. Different slopes $\rightarrow$ exactly one intersection point.

C) Two distinct lines intersect at most once.

D) Would require the same line (same slope AND same intercept).

■ **TRAP ALERT:** Different slopes = exactly one solution. Same slope = parallel (0 or ∞ solutions).

■ **KEY INSIGHT:** Two lines: different slopes $\rightarrow$ 1 solution. Same slope, different intercept $\rightarrow$ 0. Same line $\rightarrow \infty$.

■ **FAST METHOD:** Slopes 2 and -1 differ $\rightarrow$ exactly 1 solution.

Q4. Correct Answer: B [EASY]

STEP-BY-STEP SOLUTION:

Equation 1: $ax+2y=6$. Equation 2: $3x+y=4$.

For NO solution: coefficient ratios must match but constant ratio must differ.

Ratio of y-coefficients: $2/1=2$. So we need ratio of x-coefficients = 2: $a/3=2 \rightarrow a=6$.

Check constant ratio: $6/4=3/2 \neq 2$. ✓ So $a=6$ gives parallel lines (no solution). ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $a/3=1 \neq 2/1=2$. Different slope ratios $\rightarrow$ exactly 1 solution.

B) ✓ CORRECT — CORRECT. $a/3=2=2/1$ (slopes match) but $6/4 \neq 2$ (constants differ) $\rightarrow$ 0 solutions.

C) $a/3=0.5 \neq 2$. Different slopes $\rightarrow$ 1 solution.

D) $a/3=-2 \neq 2$. Different slopes $\rightarrow$ 1 solution.

■ **TRAP ALERT:** For no solution: x-coeff ratio = y-coeff ratio $\neq$ constant ratio. All three ratios matter.

■ **KEY INSIGHT:** Set $a/3 = 2/1 \rightarrow a=6$. Then verify $6/4 \neq 2$ (constant ratio differs) $\rightarrow$ no solution confirmed.

■ **FAST METHOD:** $a/3=2/1 \rightarrow a=6$. Constant ratio $6/4=1.5 \neq 2 \rightarrow$ parallel $\rightarrow$ no solution.

Q5. Correct Answer: D [MEDIUM]

STEP-BY-STEP SOLUTION:

Eq1: $4x-6y=10 \rightarrow$ divide by 2: $2x-3y=5$.

Eq2: $2x-3y=k$.

For NO solution: same slope but different constants $\rightarrow k \neq 5$.

The question says the system has no solution — so ANY value $k \neq 5$ makes it have no solution.

Answer: D (any value except 5). ✓

Note: $k=5$ would make both equations $2x-3y=5 \rightarrow$ same line $\rightarrow \infty$ solutions.

WHY EACH OPTION IS / IS NOT CORRECT:

A) $k=5$: both equations become $2x-3y=5 \rightarrow$ SAME LINE $\rightarrow \infty$ solutions. NOT no solution.

B) $k=4 \neq 5$: $2x-3y=4$ parallel to $2x-3y=5 \rightarrow 0$ solutions. ✓ But D is more complete.

C) $k=6 \neq 5$: also gives 0 solutions. But D is more complete.

D) ✓ **CORRECT — CORRECT.** For any $k \neq 5$, the system has no solution.

■ **TRAP ALERT:** $k=5$ gives INFINITE solutions (same line), NOT no solution. This is the classic trap.

■ **KEY INSIGHT:** When coefficient ratios match, the only question is whether the constants also match (∞ sol.) or not (0 sol.).

■ **FAST METHOD:** $\text{Eq1} \div 2 \rightarrow 2x-3y=5$. For no solution with $2x-3y=k$: $k \neq 5$.

Q6. Correct Answer: C [MEDIUM]

STEP-BY-STEP SOLUTION:

Substitute $y=-x+3$ into $x^2+y=5$:

$x^2+(-x+3)=5 \rightarrow x^2-x+3=5 \rightarrow x^2-x-2=0$.

Factor: $(x-2)(x+1)=0 \rightarrow x=2$ or $x=-1$.

Both are real solutions $\rightarrow 2$ intersection points. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Discriminant = $1+8=9>0 \rightarrow$ two real solutions.

B) One solution would require discriminant=0.

C) ✓ **CORRECT — CORRECT.** $x^2-x-2=0$ has discriminant $9>0 \rightarrow 2$ solutions.

D) Nonlinear systems can't have infinitely many solutions (generally).

■ **TRAP ALERT:** For nonlinear systems: substitute, form a polynomial equation, and count real roots using discriminant.

■ **KEY INSIGHT:** The number of intersections = number of real solutions of the resulting equation after substitution.

■ **FAST METHOD:** $x^2-x-2=0 \rightarrow \text{disc}=1+8=9>0 \rightarrow 2 \text{ real roots} \rightarrow 2 \text{ intersection points.}$

Q7. Correct Answer: C [MEDIUM]

STEP-BY-STEP SOLUTION:

Eq1: $2x+ky=8$. Eq2: $6x+12y=24 \rightarrow \text{divide by 3: } 2x+4y=8$.

For ∞ solutions: Eq1 must equal simplified Eq2.

Comparing $2x+ky=8$ with $2x+4y=8$: $k=4$. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $k=2$: $2x+2y=8 \neq 2x+4y=8$. Parallel $\rightarrow 0$ solutions.

B) $k=3$: $2x+3y=8 \neq 2x+4y=8$. Parallel $\rightarrow 0$ solutions.

C) ✓ **CORRECT — CORRECT.** $k=4$: both equations become $2x+4y=8 \rightarrow \text{same line} \rightarrow \infty$ solutions.

D) $k=6$: $2x+6y=8 \neq 2x+4y=8$. Parallel $\rightarrow 0$ solutions.

■ **TRAP ALERT:** Simplify Eq2 first ($\div 3$) before comparing. k must match the simplified y -coefficient (4, not 12).

■ **KEY INSIGHT:** Always simplify equations to lowest terms before applying the infinitely-many-solutions condition.

■ **FAST METHOD:** $\text{Eq2} \div 3$: $2x+4y=8$. Match Eq1: $k=4$.

Q8. Correct Answer: B [MEDIUM]

STEP-BY-STEP SOLUTION:

Eq1: $px + 4y = 12$. Eq2: $x + 2y = 6$.

Multiply Eq2 by 2: $2x + 4y = 12$.

For infinitely many solutions, Eq1 must be identical to $2x + 4y = 12$.

Comparing: $px = 2x \rightarrow p = 2$. ✓

When $p=2$: both equations reduce to $x + 2y = 6 \rightarrow \text{same line} \rightarrow \infty$ solutions.

WHY EACH OPTION IS / IS NOT CORRECT:

A) $1x+4y=12 \rightarrow x+4y=12 \neq x+2y=6$. Different y -coeffs $\rightarrow 1$ solution.

B) ✓ **CORRECT — CORRECT.** $2x+4y=12 = 2(x+2y=6)$. Same line $\rightarrow \infty$ solutions.

C) $3x+4y=12 \neq 2x+4y=12$. Different x -coeffs $\rightarrow 1$ solution.

D) $4x+4y=12 \rightarrow x+y=3 \neq x+2y=6$. Different slopes $\rightarrow 1$ solution.

■ **TRAP ALERT:** Simplify Eq2 by multiplying to match Eq1's structure, then compare coefficients.

■ **KEY INSIGHT:** For ∞ solutions: Eq1 must be a scalar multiple of Eq2. Multiply Eq2 by 2 $\rightarrow$ compare with Eq1.

■ **FAST METHOD:** $\text{Eq2} \times 2: 2x+4y=12$. Match $\text{Eq1}: p=2$.

Q9. Correct Answer: C [MEDIUM]

STEP-BY-STEP SOLUTION:

Substitute $y=2x-1$ into $y=x^2-4$:

$$2x-1=x^2-4 \rightarrow x^2-2x-3=0.$$

Factor: $(x-3)(x+1)=0 \rightarrow x=3$ or $x=-1$.

Both are real $\rightarrow 2$ intersection points. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Discriminant $=4+12=16>0 \rightarrow$ solutions exist.

B) One solution would require discriminant $=0$.

C) ✓ CORRECT — CORRECT. $x^2-2x-3=0 \rightarrow \text{disc}=16>0 \rightarrow 2$ real solutions.

D) A line and parabola intersect at most twice.

■ **TRAP ALERT:** A line and a parabola can intersect at 0, 1, or 2 points — never 3.

■ **KEY INSIGHT:** Substitute the linear equation into the quadratic $\rightarrow$ solve the resulting quadratic for number of real roots.

■ **FAST METHOD:** $x^2-2x-3=0 \rightarrow (x-3)(x+1)=0 \rightarrow 2$ solutions.

Q10. Correct Answer: C [HARD]

STEP-BY-STEP SOLUTION:

System: $cx + 2y = 4 \dots (1)$ and $2x + cy = 6 \dots (2)$.

For NO solution: the determinant must be 0 AND the system must be inconsistent.

$$\det = c \cdot c - 2 \cdot 2 = c^2 - 4.$$

$$\det = 0 \rightarrow c^2 = 4 \rightarrow c = 2 \text{ or } c = -2.$$

Check $c = 2$: $\text{Eq1} \rightarrow 2x+2y=4 \rightarrow x+y=2$. $\text{Eq2} \rightarrow 2x+2y=6 \rightarrow x+y=3$.

$x+y=2$ and $x+y=3 \rightarrow$ parallel, different constants $\rightarrow$ NO solution. ✓

Check $c = -2$: $\text{Eq1} \rightarrow -2x+2y=4 \rightarrow -x+y=2$. $\text{Eq2} \rightarrow 2x-2y=6 \rightarrow x-y=3$.

Rewrite Eq2 : $-x+y=-3$. Compare Eq1 : $-x+y=2$. Same slope, different constant $\rightarrow$ NO solution. ✓

Both $c=2$ and $c=-2$ give no solution. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) $c=2$ works, but $c=-2$ also gives no solution.

B) $c=-2$ works, but $c=2$ also gives no solution.

C) ✓ CORRECT — CORRECT. Both values give $\det=0$ with inconsistent (parallel) lines.

D) $\det=16-4=12 \neq 0 \rightarrow$ unique solution, not no solution.

■ **TRAP ALERT:** $c^2=4$ has TWO solutions ($c=\pm 2$), both of which give parallel lines. Don't stop at one.

■ **KEY INSIGHT:** For no solution: $\det=0$ (parallel slopes) AND check that constants differ (not same line).

■ **FAST METHOD:** $c^2=4 \rightarrow c=\pm 2$. Both give parallel, inconsistent lines $\rightarrow$ no solution.

Q11. Correct Answer: C [HARD]

STEP-BY-STEP SOLUTION:

System: $3x^2-3y=9$ (using $k=3$) and $y=3x-3$.

Substitute: $3x^2-3(3x-3)=9 \rightarrow 3x^2-9x+9=9 \rightarrow 3x^2-9x=0 \rightarrow 3x(x-3)=0$.

$x=0$ or $x=3 \rightarrow 2$ solutions. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Both $x=0$ and $x=3$ are valid real solutions.

B) Two solutions exist, not one.

C) ✓ CORRECT — CORRECT. $3x(x-3)=0$ gives $x=0$ and $x=3 \rightarrow 2$ solutions.

D) A parabola and line can share at most 2 points.

■ **TRAP ALERT:** 'Tangent' applies when discriminant=0. Here with $k=3$, there are 2 distinct intersection points.

■ **KEY INSIGHT:** Substitute the linear equation into the quadratic, simplify, and count real roots.

■ **FAST METHOD:** $3x^2-9x=0 \rightarrow 3x(x-3)=0 \rightarrow x=0$ or $x=3 \rightarrow 2$ solutions.

Q12. Correct Answer: D [HARD]

STEP-BY-STEP SOLUTION:

$x^2 + y^2 = 9$ is a circle of radius 3 centred at the origin.

$x + y = c$ is a line with slope -1 and y-intercept c .

For the line to intersect the circle: distance from centre to line $\leq$ radius.

Distance from (0,0) to line $x + y - c = 0$: $d = |c|/\sqrt{(1^2+1^2)} = |c|/\sqrt{2}$.

Set $d \leq 3$: $|c|/\sqrt{2} \leq 3 \rightarrow |c| \leq 3\sqrt{2} \approx 4.24$.

Verify $c = \pm 4$: $|4|/\sqrt{2} = 2\sqrt{2} \approx 2.83 < 3$ ✓ $\rightarrow$ line intersects circle.

Verify $c = \pm 5$: $|5|/\sqrt{2} \approx 3.54 > 3 \rightarrow$ line does NOT intersect circle.

Integer values of c with $|c| \leq 4.24$: $c = -4, -3, -2, -1, 0, 1, 2, 3, 4 \rightarrow 9$ values. ✓

WHY EACH OPTION IS / IS NOT CORRECT:

A) Too few — misses many valid c values.

B) Only counts $|c| \leq 2$, which is too restrictive.

C) Only counts $|c| \leq 3$. But $c = \pm 4$ also works since $4/\sqrt{2} < 3$.

D) ✓ CORRECT — CORRECT. Integers from -4 to 4 inclusive = 9 values. All satisfy $|c|/\sqrt{2} \leq 3$.

■ **TRAP ALERT:** $3\sqrt{2} \approx 4.24$, not 3. Don't round down to 3; $c = \pm 4$ is valid because $4 < 4.24$.

■ **KEY INSIGHT:** Distance formula for point to line: $d = |Ax + By + C|/\sqrt{A^2 + B^2}$. Here $d = |c|/\sqrt{2} \leq 3$.

■ **FAST METHOD:** $|c| \leq 3\sqrt{2} \approx 4.24$. Largest integer: 4. Range: -4 to 4 $\rightarrow$ 9 integers.

TOPIC SUMMARY — SYSTEMS OF EQUATIONS — NUMBER OF SOLUTIONS

Digital SAT Math — Comprehensive Review

■ COMMON MISTAKES

■ Confusing 'no solution' with 'infinite solutions': No solution $\rightarrow$ same slope, DIFFERENT intercept. Infinite solutions $\rightarrow$ SAME line (all ratios equal).

■ For k that makes parallel lines: k makes lines the same line (∞ sol.), while any other k gives no solution. Never confuse which direction.

■ For nonlinear systems: the number of solutions equals the number of real roots of the substituted polynomial — always check the discriminant.

■ MUST REMEMBER

■ Three cases — coefficients $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$: $a_1/a_2 = b_1/b_2 = c_1/c_2$ (∞ solutions); $a_1/a_2 = b_1/b_2 \neq c_1/c_2$ (0 solutions); $a_1/a_2 \neq b_1/b_2$ (1 solution).

■ Always simplify equations before comparing ratios (divide by GCF).

■ For circle–line systems: use the distance formula from centre to line to determine the number of intersections.

■ Determinant of a 2×2 system: $\det = a_1b_2 - a_2b_1$. $\det \neq 0 \rightarrow$ unique solution; $\det = 0 \rightarrow 0$ or ∞ .

■ EXPECTED ON DSAT

■ Number-of-solutions questions for LINEAR systems appear on nearly every test, especially in Hard modules.

■ The 'which value of k gives no solution vs infinite solutions' question type is among the most frequently tested at Medium–Hard level.

■ Nonlinear systems (line + parabola, line + circle) appear at Hard level — substitute and count real roots.